

Bayesian Inference

INS-605: Data Analysis II





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Helpful Resources

🌐 Understanding the Difference between Bayesian and Frequentist Statistics by Fornacon-Wood et al. (2021).



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Understanding the Differences Between Bayesian and Frequentist Statistics



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Learning Objectives

By the end of this lecture, you should be able to:

- Explain the motivation for **Bayesian inference**.
- Distinguish **Frequentist** and **Bayesian** interpretations.
- Apply **Bayes' theorem** to estimate parameters.
- Explain the roles of the *prior*, *likelihood*, and *posterior*.
- Interpret **Bayesian credible intervals**.
- Perform Bayesian updating using a **Beta-Binomial model**.
- Understand **posterior predictive inference**.

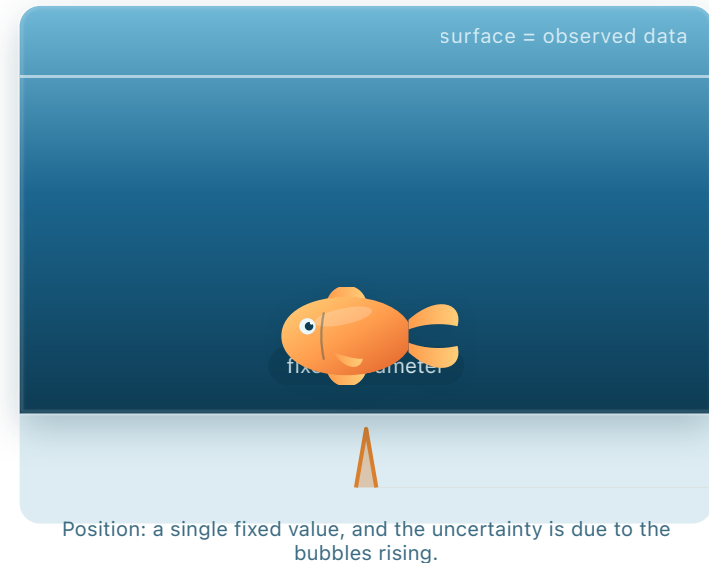
1 Motivation

1.1 Frequentist Inference

- Frequentist framework:
 - The parameter is **fixed**.
 - The parameter is **unknown**.
 - The **data are random**.
- Example: Eggs survival model:
 $X_1, \dots, X_n \sim \mathcal{B}(p)$, p is **unknown**.
- **Frequentist inference**:
 - **Point estimate**: find $\hat{p}$, for example by maximizing the likelihood function.
 - **Interval estimate**: find an interval $I_{1-\alpha} = [a(\hat{p}), b(\hat{p})]$ with $\mathbb{P}(p \in I_{1-\alpha}) = 1 - \alpha$.

Frequentist

The parameter is **fixed** and **unknown**. The fish doesn't move — we just observe what surfaces.



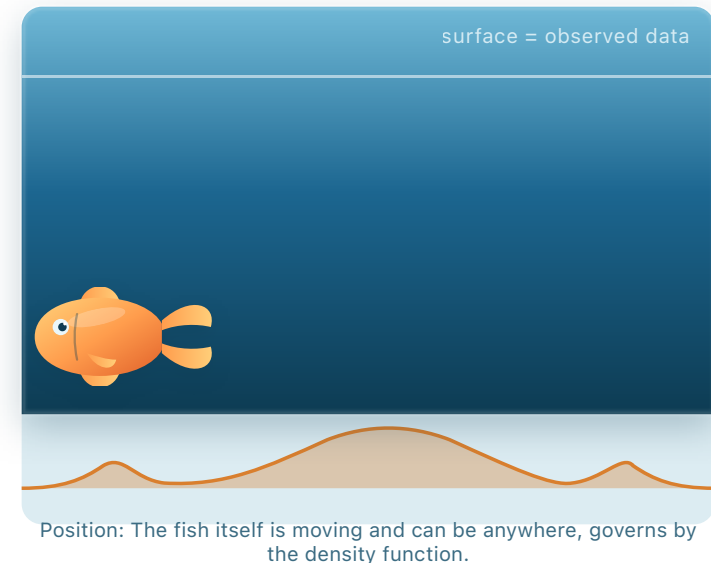
1.2 Bayesian Thinking

- The parameter is **uncertain** & data is **uncertain** but **observed**.
- Ex: Before observing the data, you may believe that it's fair ($p = 0.5$). However, after some tosses, you may change your belief about the parameter p .



Bayesian

The parameter itself has a **distribution**. The fish swims and leaves its own trail of data behind.



- **Prior:** Your initial belief in value of parameter.
- **Bubbles:** Data $X_1, X_2, \dots$ are observed.
- **Posterior:** Your updated belief in value of parameter after observing the data.

Frequentist Vs Bayesian

Question	Frequentist	Bayesian
What is the parameter?	Unknown and <i>fixed</i>	Unknown and <i>uncertain</i>
What is random?	Data	Parameter uncertainty → data
Main tool	Likelihood	Bayes' theorem
Main result	Point estimate (MLE)	Posterior distribution
Interval	Confidence interval	Credible interval
Prediction	Based on estimated parameters	Based on posterior uncertainty

2 Bayes' Theorem

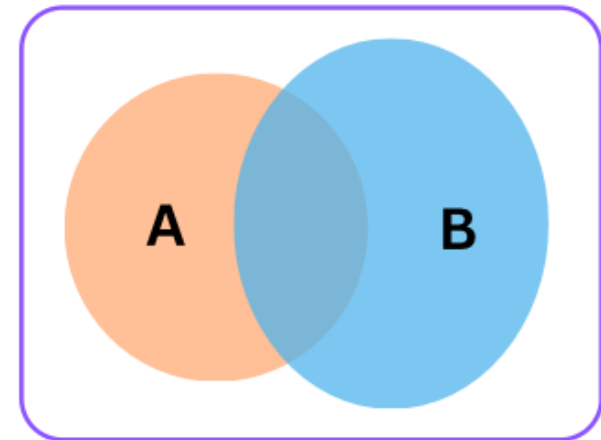
2.1 Bayes' Theorem For Events

- For two events A and B with non-zero probabilities, Bayes' theorem states:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

where:

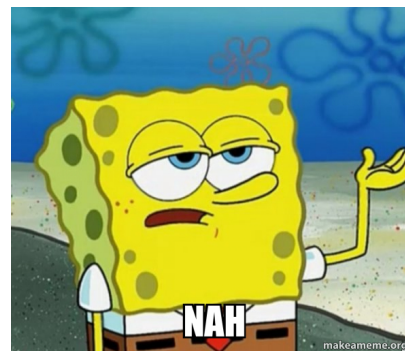
- $P(A)$: prior probability
- $P(B | A)$: likelihood
- $P(B)$: evidence
- $P(A | B)$: posterior probability



Example: Disease Testing

- Suppose a disease affects 50% of the population ($D = 1$).
 - A test is **99% accurate**:
 - Correctly identifies 99% of those with the disease,
 - Correctly identifies 99% of those without the disease.
 - If a person tests positive, how likely do you think they actually have the disease?
- $P(T = 1 \mid D = 1) = 0.99$.
 - $P(D = 1) = 0.5$.
 - Testing positive:
 $P(T = 1)$
 $= 0.99 \times 0.5 + 0.01 \times 0.5$
 $= 0.5$ (why?)
 - Hence, $P(D = 1 \mid T = 1) = 0.99$.
 - Is this surprising?

$$P(D = 1 \mid T = 1) = \frac{P(T = 1 \mid D = 1)P(D = 1)}{P(T = 1)}.$$



Example: Rare Disease Testing

- Suppose a disease affects **only 1%** of the population.

- $P(T = 1 \mid D = 1) = 0.99$.

- $P(D = 1) = 0.01$.

- Testing positive:

$$P(T = 1)$$

$$= 0.99 \times 0.01 + 0.01 \times 0.99$$

$$= 0.0198 \text{ (why?)}$$

- Hence,

$$P(D = 1 \mid T = 1) = \frac{0.99 \times 0.01}{0.0198} = 0.5.$$

- Is this surprising?

- If there are 1M people, then

Test	Sick	Healthy
Positive	9,900	9,900
Negative	100	980,100
Total	10,000	990,000

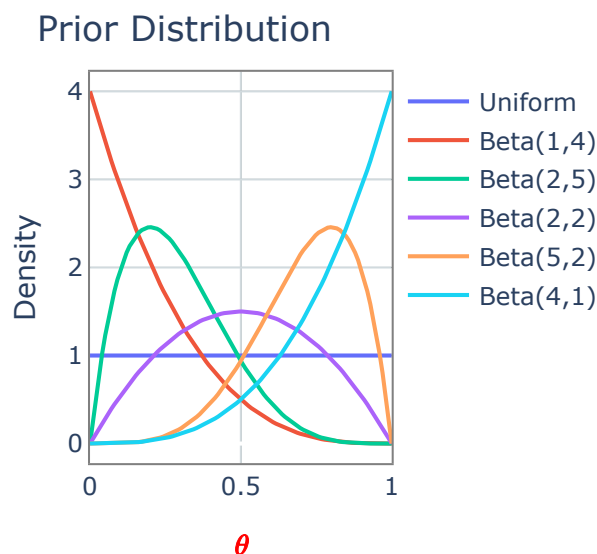
- For very rare diseases, even a highly accurate test can produce a large number of **false positives (9,900)**.
- A second test can help reduce **false positives**.

2.2 Bayes' Theorem For Parameters

- Let x be the observed data and θ be an unknown parameter.

$$\underbrace{p(\theta | x)}_{\text{Posterior}} = \frac{\overbrace{p(x | \theta)}^{\text{Likelihood}} \overbrace{p(\theta)}^{\text{Prior}}}{\underbrace{p(x)}_{\text{How likely to observe data } x}} \propto p(x | \theta)p(\theta).$$

- Coin toss** example, let $\theta = p$ be unknown probability of head per toss (**success**).
 - Prior:** $\theta \sim \text{Beta}(2, 2)$, i.e., $f(\theta) = \frac{\theta(1-\theta)}{B(2,2)}$.
 - Data:** $N = 30$ and $n(H) = 18$.
 - Likelihood:** $p(x = 18 | \theta) = \binom{30}{18} \theta^{18} (1 - \theta)^{12}$.

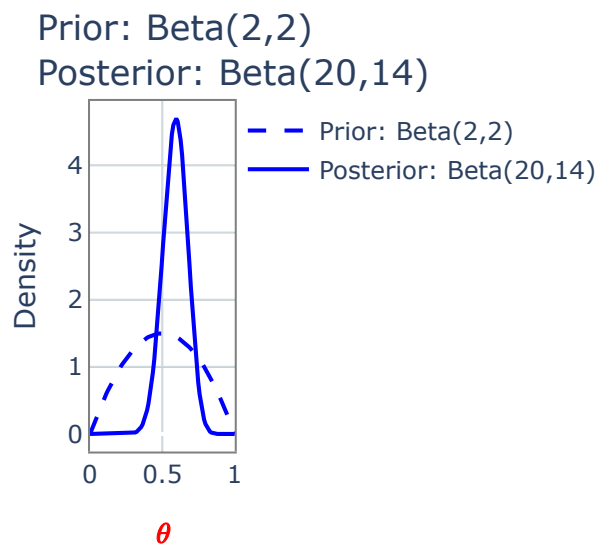


2.2 Bayes' Theorem For Parameters

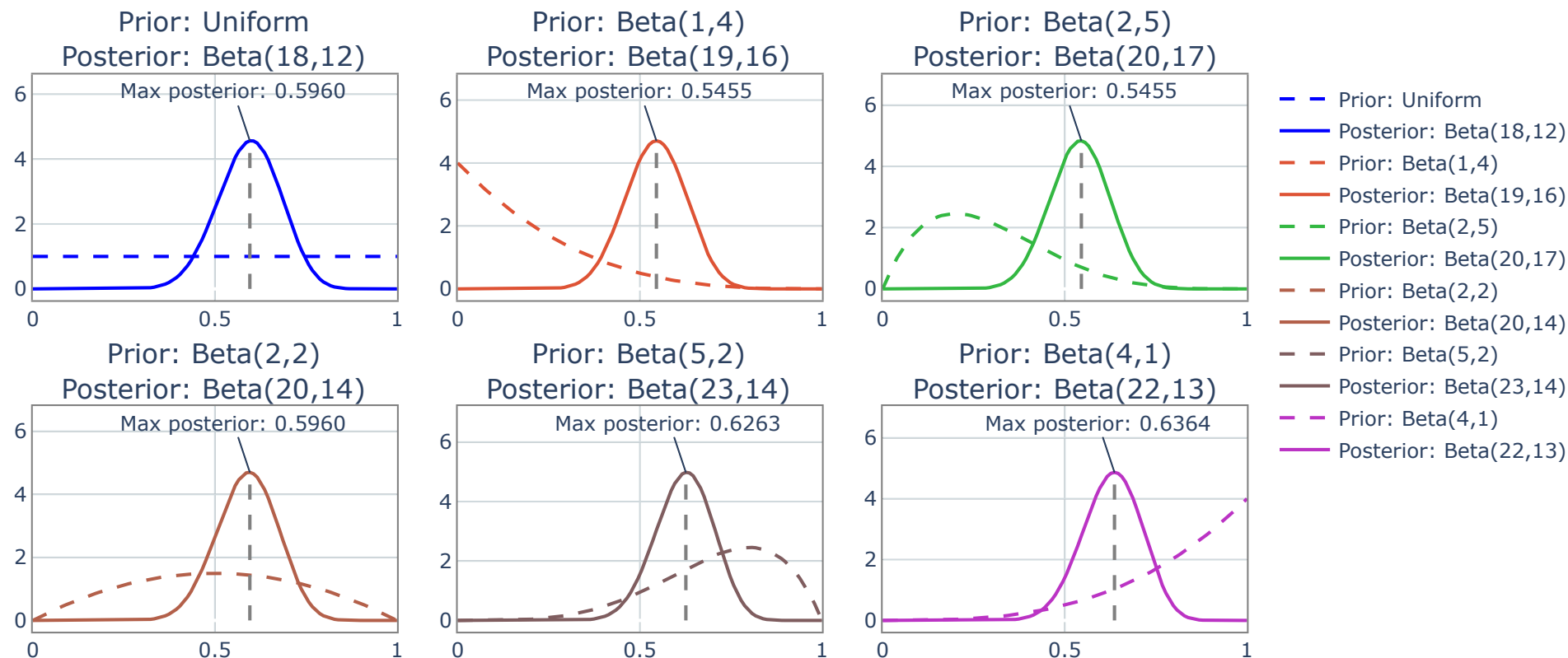
- Let x be the observed data and θ be an unknown parameter.

$$\underbrace{p(\theta | x)}_{\text{Posterior}} = \frac{\overbrace{p(x | \theta)}^{\text{Likelihood}} \overbrace{p(\theta)}^{\text{Prior}}}{\underbrace{p(x)}_{\text{How likely to observe data } x}} \propto p(x | \theta)p(\theta).$$

$$\begin{aligned} p(\theta | x = 18) &\propto p(x = 18 | \theta)p(\theta) \\ &= \left[\binom{30}{18} \theta^{18} (1 - \theta)^{12} \right] [\theta(1 - \theta)] \\ &\propto \theta^{19} (1 - \theta)^{13} \\ &= \theta^{20-1} (1 - \theta)^{14-1} \\ \Rightarrow \underbrace{(\theta | x = 18)}_{\text{New belief of } \theta} &\sim \text{Beta}(20, 14) \quad (\text{Posterior}). \end{aligned}$$



Influence Of Prior

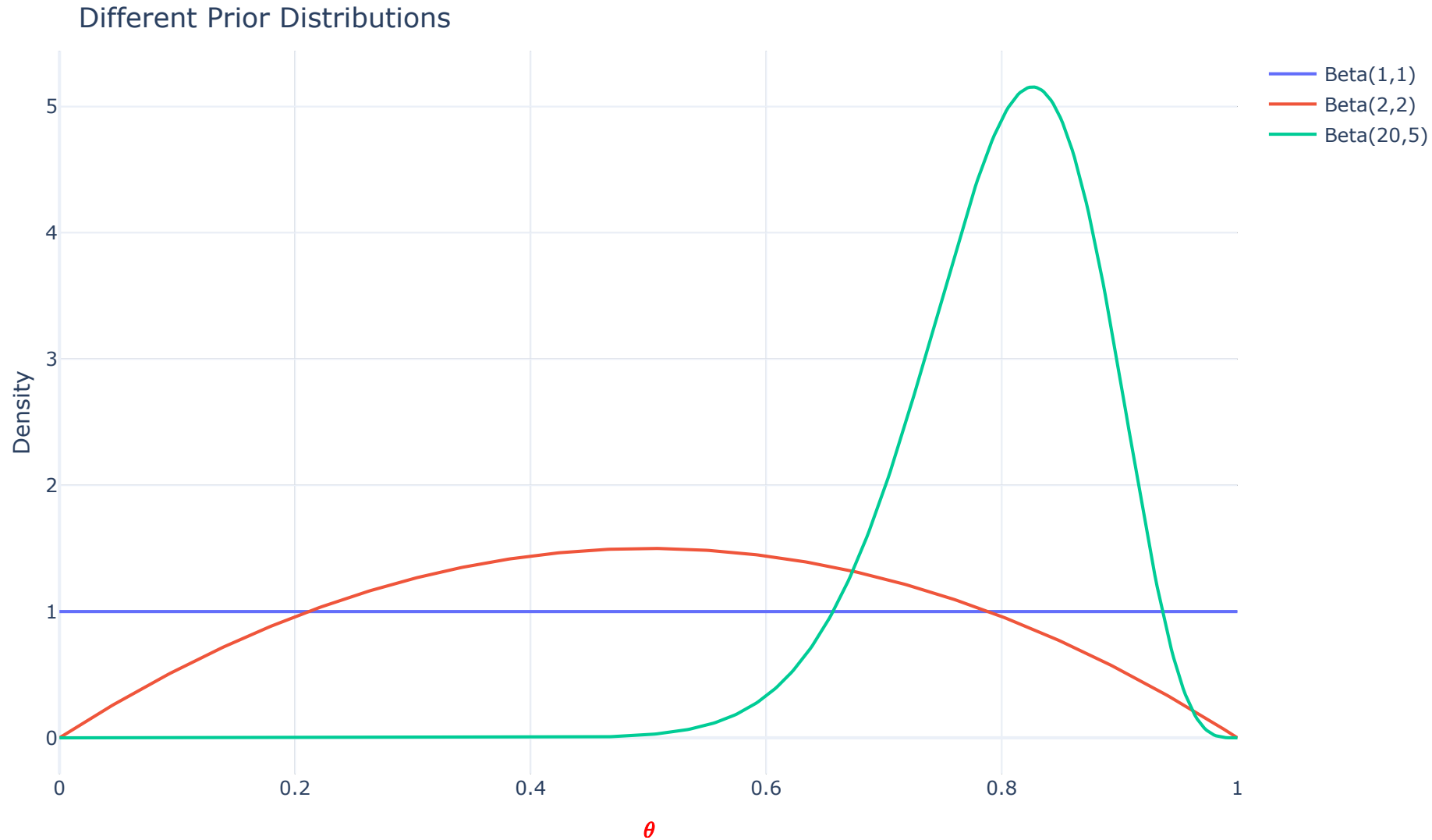


- **Prior belief** is expressed by $p(\theta)$ (dashed lines), and when observing data $\mathbf{x}$, it updates to a new belief (**posterior**) $p(\theta | \mathbf{x})$.
- Different priors can result in different posterior distributions.

Weak And Informative Priors

- A prior can be:
 - Weakly or strongly informative.
 - Symmetric or asymmetric.
 - Based on previous studies or expert knowledge.
 - Chosen for mathematical convenience.
- For example: $\theta \sim \text{Beta}(1, 1)$ is uniform on $U[0, 1]$.
- Meanwhile: $\theta \sim \text{Beta}(20, 5)$ expresses a stronger belief that θ is relatively large.

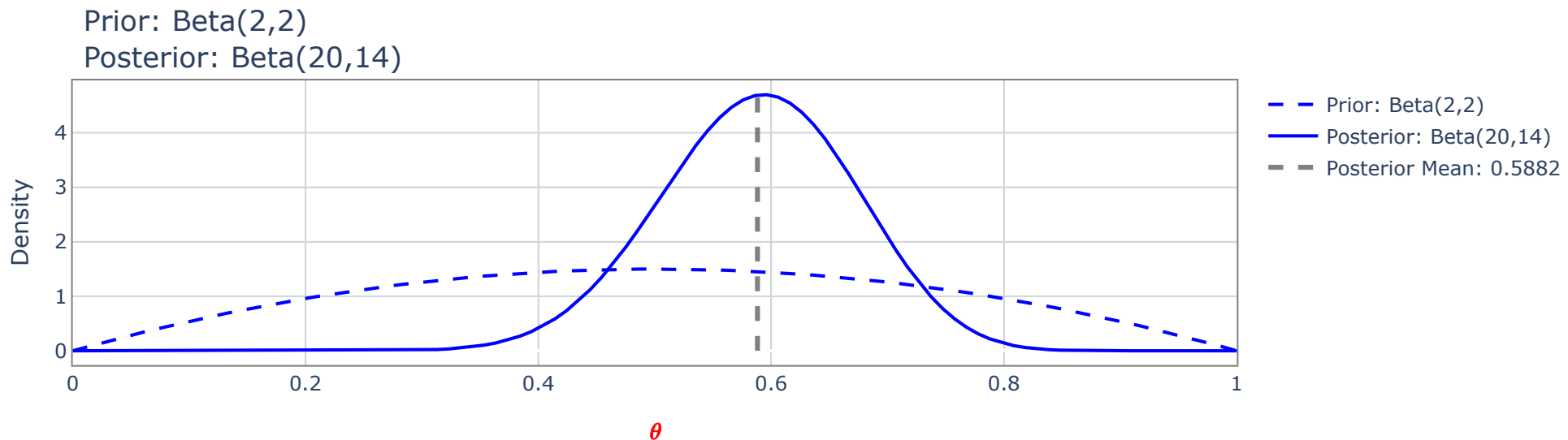
Python: Visualizing Beta Priors



3 Posterior Mean

3.1 Posterior Mean

- In **Bayesian Framework**, **Posterior Mean**, the expected value of the parameter given the observed data, i.e., $\mathbb{E}[\theta|x]$, is a point summary of the parameter of the population.
- After computing the posterior, we compute its expectation.
 - For example if $(\theta|x) \sim \text{Beta}(\alpha, \beta) \Rightarrow \mathbb{E}[\theta|x] = \frac{\alpha}{\alpha+\beta}$.



3.2 MLE Vs Posterior Mean

- Let's compare **MLE** and **Posterior Mean** on **Coin Toss** example.
- **Maximum Likelihood Estimator (MLE):**
 - Log-likelihood: $\ell(\theta) = 18 \ln(\theta) + 12 \ln(1 - \theta) + \text{cont.}$
 - $\ell'(\theta) = 18/\theta - 12/(1 - \theta)$
 - **MLE:** $\hat{\theta} = 18/30 = 0.6$.
 - This is purely data-driven.
- **Posterior Mean:**
 - For prior $\theta \sim \text{Beta}(2, 2)$
 - Posterior $(\theta \mid \mathbf{x}) \sim \text{Beta}(20, 14)$
 - **Posterior Mean:** $\hat{\theta} = \mathbb{E}(\theta \mid \mathbf{x}) = \frac{\alpha}{\alpha + \beta} = \frac{20}{20 + 14} = 0.5882$.

4 Credible Intervals

4.1 Credible Intervals

- **Posterior mean** is point summary of the **parameter**.
- A credible interval is an interval estimate of a parameter that captures the uncertainty in the parameter.
- If $\alpha \in (0, 1)$, be a level of error, then a credible interval at level $1 - \alpha$ of θ denoted by $CI_{1-\alpha} = [a, b]$ such that $\mathbb{P}(a \leq \theta \leq b \mid x) = 1 - \alpha$.
- This means:

Given the observed data x and the prior $p(\theta)$, there is a $1 - \alpha$ posterior probability that $\theta \in [a, b]$.

- The interval is calculated from the posterior distribution.

Bayesian Prediction

- Bayesian inference can also predict a future observation.
- Suppose we want to predict the chance of getting head next:

$$\begin{aligned} p(X_{new} = 1 \mid \mathbf{x}) &= \int_0^1 \underbrace{p(X_{new} = 1 \mid \theta)}_{\text{Chance of head}} p(\theta \mid \mathbf{x}) d\theta \\ &= \int_0^1 \theta p(\theta \mid \mathbf{x}) d\theta \\ &= \mathbb{E}(\theta \mid \mathbf{x}) = 20/34 = 0.5882. \end{aligned}$$

- **Remark** Bayesian theory fits well with online learning as Posterior can be considered a new Prior for the next observation.

Frequentist Vs Bayesian Workflow

- **Frequentist:**

```
1 Choose probability model
2     ↓
3 Construct likelihood
4     ↓
5 Optimize likelihood
6     ↓
7 Obtain MLE
8     ↓
9 Construct confidence interval
10    ↓
11 Make predictions
```

- **Bayesian:**

```
1 Choose probability model
2     ↓
3 Specify prior distribution
4     ↓
5 Construct likelihood
6     ↓
7 Combine prior and likelihood
8     ↓
9 Obtain posterior distribution
10    ↓
11 Calculate posterior summaries
12    ↓
13 Construct credible intervals
14    ↓
15 Make posterior predictions
```

Conjugate Priors

Likelihood	Prior	Posterior
Bernoulli / Binomial	Beta	Beta
Poisson	Gamma	Gamma
Normal mean with known variance	Normal	Normal
Multinomial	Dirichlet	Dirichlet

- Conjugacy makes Bayesian inference mathematically convenient.
- Many real-world models do not have a closed-form posterior.
 - Logistic regression
 - Neural network models...

🥳 Yeahhhh 🍷 !!!



Let's take a break!